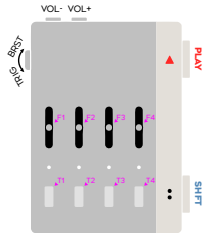


Wakes v0.7.0 - Pg.1 - Welcome



• Hey! I'm Adara. Welcome to **Wakes**, a custom firmware for Teenage Engineering's unreleased "SP-1" stem player. Wakes adapts code from three Eurorack modules into one device: **Plaits**, a macro-synthesizer, **Marbles**, a playful random sequencer, and **Yarns**, a signal converter. The modules were created by **Émilie Gillet**, otherwise known as **Mutable Instruments**.

• First and foremost, you should know how to turn the device on or off. Holding down **SHFT** for 1.5 seconds will show a brief animation before the device will turn on, going to its first page, **Plaits**.

• While turned on, holding **SHFT** for 3 seconds will turn the device off.

• While plugged in, Wakes has a standby mode.

• While on, any input while **SHFT** is held down will interrupt the shutdown behaviour, allowing you to work inside **SHFT** pages.

• Even if interrupted, holding down **SHFT** for 25 seconds will cause the face LEDs to flash quickly, notifying you that the SP-1 is about to perform an emergency shutdown in 5 seconds if you don't let **SHFT** go. Release and return to holding **SHFT** to get back to what you were doing. This is intended as an emergency backstop.

• At any point, even when off, tapping **SHFT** for a short moment will bring up a battery meter.

• While using **Wakes**, you'll be bouncing between **Plaits** to design sounds, and **Marbles** to sequence it.

• As you bounce back and forth, the face LED's will show you what's going on through a system of **glyphs**.

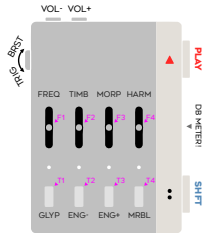
• Faders **F1-F4** have a catch-up mechanism with the actual digital values, so moving them around and swapping pages doesn't create unexpected behaviours. For clarity, follow the LED's to know where the set values reside.

• Each module has a **SHFT** page containing advanced parameters and options that open up avenues for experimenting.

• Double-tapping **SHFT** enters a deep tinkering page called **SETTINGS**. There are two **SETTINGS** pages: One for **Plaits**, one for **Marbles**.

• Legend: ○ - Lit LED ● - Unlit LED





- **Wakes'** first page is **Plaits**: A synthesizer containing a multitude of sound engines representing different sound design techniques, with parameters attached to explorative macros. A macro-synthesizer.

- **Plaits** has four main faders: **FREQ**, **TIMB**, **MORP** and **HARM**. Each of these controls several components of the selected synthesis engine.

- **Plaits** has a built-in lowpass-gate and amplifier, controlled by an internally-routed **decay envelope**.

- For now, you can read about what the main parameters do for each engine in commonly available **Plaits** guides.

- While in **Plaits**, the four side LED's will show a dB meter. Each LED represents 6dB of dynamic range.

- Generally speaking, **FREQ** will control the overall pitch of the sound. **TIMB** sweeps the spectral content from dark/sparse to bright/dense. **MORP** explores lateral timbral variations. **HARM** controls the frequency spread/overall balance of the spectrum.

- The rocker switch plays notes, triggering the decay envelope on the filter/amp. Press it down to send a single trigger, pull it up to send a burst.

- Engines are known by glyphs. Pressing **GLYP** will flash the glyph representing the selected engine. **ENG+** for the next one, **ENG-** to go back.

- **PLAY** engages **Marbles'** clock, which is patched by default to trigger **Plaits** and give it musical notes.

- Press **MRBL** to go to **Marbles**.

- Hold **SHIFT** to go to **Plaits SHIFT**:

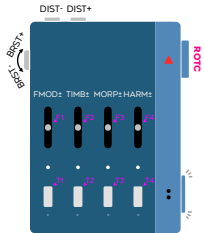
- Each fader here represents a modulation depth control for incoming control signals. By default, **Plaits'** **decay envelope** is assigned to all of the four parameters. Use the faders to modulate **Plaits'** macros! Fun!

- The rocker switch selects a rhythmic subdivision for **BRST'** retriggers.

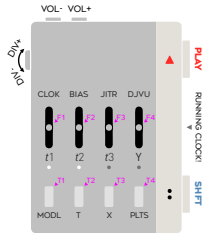
- **DIST-** and **DIST+** control the depth of a master softclipper.

- Holding down any of **T1-T4** for 2 seconds **Unpatches** **Marbles** modulators from that parameter.

- Holding down **PLAY** for 3 seconds **Rips Out Plaits' Cables**, returning it to its default configuration.



Wakes v0.7.0 - Pg.3 - Marbles - t Side (left)

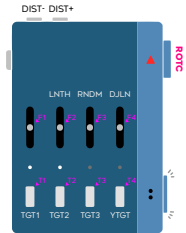


- Marbles has two sides: **t** and **X**. The glyphs will show you which side you are on: ○○○○ for **t** and ●●●● for **X**. Each side has 3 outputs: **t1-t3**, **X1-X3**. An extra output **Y** is offered. It's weird. :)
- **t** is a random gate generator. It emulates a player following a stable metronome. **CLOK** determines the BPM (default: 120). **BIAS** determines the distribution of gate signals depending on the selected model using **MODL**. **JITR** changes the player's accuracy. stable > chaotic.
- **DJVU** controls the probability of reusing played patterns. At center, it is a locked loop. Moving down adds more unlocked randomness. Up from centre, memorized notes are randomized instead. **DJLN** sets loop length.

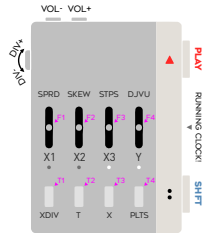
- **t2** is always a stable clock. It's a 50/50 gate, 50% on, 50% off.
- Pressing **MODL** swaps between a list of distribution models for **t1** and **t3**:
 - - coin toss: gates are evenly distributed.
 - - clusters: random pairs of clock subdivisions appear at each side.
 - - drums: Steps through drum pattern tables containing rhythms.
 - - independent coin: each side tosses its own. **t1** and **t3** can coincide.
 - - divider: **BIAS** selects from fixed subdivision pairs. 8:1 > 1:1 > 1:8.
 - - three states: a three-way choice per tick: **t1**, **t3** or neither.
- **DIV-/DIV+** divide/multiply the clock

- Press **T3** to go to the **X** side of Marbles.
- Press **PLTS** to go back to **Plaits**.
- Hold **SHFT** to go to **t SHFT**:
 - **T1-T4** select **Plaits** modulation targets for **t1-t3**, or for **Y**. The mod targets are: ●●●●-none, ○●●●-FM, ●○○○-TIMB, ●●○○-MORP, ●○○○-HARM, ●○○○-LEVL, ●○○○-TRIG. For **Y**, ●○○○ routes to **FREQ**, contextually.
 - **LNTH** determines the length of on-states for **t1** and **t3** gate signals.
 - **RNDM** determines gate length randomization for **t1** and **t3**.
 - Holding down any of **T1-T4** for 2 seconds **Unpatches** any **Plaits** destinations.
 - Holding **PLAY** for 3 seconds **Rips Out** all of **Marbles' Cables**.

Marbles - t SHFT



Wakes v0.7.0 - Pg.4 - Marbles - X Side (right)



- **X** is a random modulation generator. Its three outputs, **X1-X3**, are clocked by the outputs of Marbles' **t** side, and generate random control signals that can vary from Sample & Hold style steps to smooth, meandering modulations. It can modulate Plaits' parameters, or quantize it to play musical intervals, creating melodic sequences, as well as enabling many other ideas.

- The fourth output, **Y**, shows up on this page as well. It is a smooth, meandering full-range random source that's clocked at 1/8th the rate of **t2**. **Y** never obeys **DJVU**, she's wild like that.

- **SPRD** spreads the concentration of control signals around the central point determined by **BIAS**.

- **SKEW** skews the signals **X1-X3** up or down in average intensity.

- **STPS** smooths or tightens the vertical steppiness of the control signals. Below-centre, the signals have random intensities. Above-centre, it quantizes **X** to a musical scale.

- **DJVU** here is the same as on the **t** side. In fact, both sections are controlled by the same memory.

- Press **XDIV** to choose the intensity diversity between the **X** outputs: identical, bump (**X2** is strongest) or tilt (**X1** is weakest, **X3** is strongest).

- Press **T2** to go back to **t**.

- Press **T4** to go back to Plaits.

Hold **SHIFT** to go to **X SHIFT**:

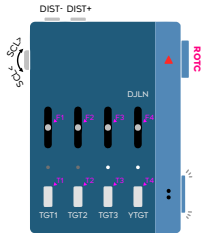
- **T1-T4** select Plaits modulation targets for **X1-X3**, or for **Y**. The mod targets are: ●●●●-none, ○●●●-FM, ●○●●-TIMB, ●●●○-MORP, ●●●○-HARM, ●○●○-FREQ, ●●○●-LEVL. **FREQ** is the default input for musical tuning. **FM** is full-range.

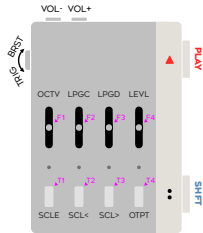
- **DJLN**, like in **t**, controls the length of the **DJVU** sequence. **DJVU** and **DJLN** are the same across the two sides.

- **SCL** select from a range of scales to quantize into, when **STPS** is above the middle point. View base available scales in the Marbles **SETTINGS** guide page.

- Holding down any of **T1-T4** for 2 seconds **Unpatches** any Plaits destinations. Hold **PLAY** to **ROTC**.

Marbles - X **SHIFT**





- Reach Plaits' **SETTINGS** page by double-tapping **SHIFT** while in the Plaits module. Single-tapping **SHIFT** will return you to where you were.

- **OCTV**'s default is its maximum value. When not maxed, it locks Plaits' available frequencies to ± 7 semitones around a chosen octave. Choose the range to play by moving the fader. Maximize it to go full-range again.

- **LPGC** lets you control the color of the low-pass gate, which is composed of a filter and amplifier. Lower values mix in more of the low-pass filter, making the internal decay envelope and **LEVL** sound duller in tone.

- **LPGD** lets you change the length of the decay envelope. Higher values set longer lengths, meaning triggered notes decay for longer when played. Any self-patching from the unpatched **SHIFT** faders will also be longer. Shorter values sound snappier and quieter overall.

- **LEVL** changes the base offset applied to Plaits' filter and amp. Moving it around fades the sound in and out. The color of this fade is affected by **LPGC**. Setting **LEVL** to maximum will effectively disable them, allowing Plaits to drone. Some Plaits engines will react beautifully to modulating **LEVL**. It's available as a mod destination for t and X. It's the default mod destination affected by Plaits' decay envelope.

- **Wakes** includes a smart Plaits quantizer feature. It lets you noodle around with notes on a scale by moving **FREQ**, without Marbles' modulation. It is **●●●●**-off by default. Activate it by pressing **SCL>**. Deactivate it by pressing **SCL<** until the glyphs show **●●●●**-off again. Once activated, it first disables any Marbles routings going into **FREQ**. Then, it takes over **FREQ** and quantizes it. Combine it with **OCTV** to lock your available notes to a short range. **SCLE** flashes the currently-active quantizer scale. Scales are the same as the ones available for Marbles, with glyphs shown in Marbles' **SETTINGS** manual page.

- Plaits features an extra output, **AUX**, containing additional content from each engine. **Wakes** gives you **OTPT** on this page to select from different output configurations: **●●●●**-OUT, **●●●●**-AUX, **●●●●**-OUT+AUX, and the **Wakes**-unique **●●●●**-OUT×AUX.

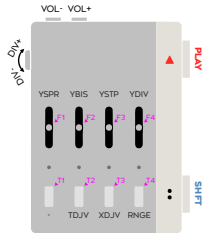
Plaits Tips

- Plaits is designed to be really breakable. Experiment with fast **BRST**, quick Marbles modulations, lots of mod targets. Just go crazy with it, and see what wilderness you end up in.

Ripping Out The Cables will take you back home.

- **FM** is a legacy Plaits input containing exp-TZFM for Plaits. It's used for atonal modulations.

- You can **TRIG** while **LEVL** is open ;)



- Reach Marbles' **SETTINGS** page by double-tapping **SHIFT** while in the Marbles module. Single-tapping **SHIFT** will return you to where you were.

- **YSPR** lets you control Y's spread, same way you do for X.

- **YBIS** lets you control Y's randomization skew towards higher or lower signal strengths.

- **YSTP** lets you control its overall steppiness. Lower values are more sinusoidal. Central values are linear, triangular. Higher values are slewed squares, up to an even sample & hold.

- **YDIV** lets you choose at which subdivision of the clock Y is clocked. This lets you accelerate it to higher values if you want.

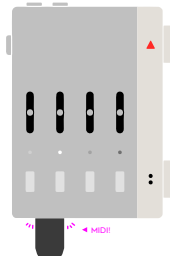
- **TDJV** and **XDJV** toggle **DJUV** for the torX sides on or off. **DJUV** is on for both by default. If you toggle it off for one of the sides, that side won't be affected by **DJUV**'s looping.

- **RNGE** sets the control signal strength range for X and Y. The options are: **0000** - 0-20%, **0001** - 0-50%, **0002** - $\pm 50\%$, **0003** - INTELLIGENT. INTELLIGENT is the default. It's a Wakes-specific upside of combining the two modules in a closed environment: It decides your voltage range selection for you, per-output, automatically, and with the target Plaits engine in mind. That said, if you still want to play around with unexpected results, the other modes remain available.

- There are 6 default scales Marbles offers for quantizing the X outputs. When you press **SCL<** or **SCL>** on Marbles' **SHIFT** page, their glyphs will flash on the face LEDs as you cycle through the list. Here they are: **0000**-major, **0001**-minor, **0002**-pentatonic, **0003**-pelog, **0004**-bhairav, **0005**-shri. If you choose one of these scales, then move **STPS** above-center, then send one of the X outputs into **FREQ**, Marbles will tell Plaits to play notes on that scale.

Marbles Tips

- You can combine multiple Marbles outputs into one Plaits destination. Using **Unpatch** disconnects only that specific output from its destination.



- Wakes supports MIDI IN and Audio OUT over the SP-1's USB-C port. Connect your SP-1 to a USB host, and it should pop up as a MIDI and USB audio device.
- Wakes automatically routes pitch and MIDI notes into **Plaits**, and routes MIDI clock into **Marbles**.
- By default, Wakes will decide control ranges and scale your CC appropriately for unipolar and bipolar parameters across the device.
- Wakes' repo includes config files: `config/midi.ini` and `config/audio.ini`. Review the files for advanced information.
- Fork the repo to build your own configurations, and customize Wakes' USB handling to your needs.

PLAITS - MIDI BEHAVIOURS

- Plaits will now receive pitch, pitchbend (± 2 default), gate, velocity (scales LEVL) and MIDI CC information from incoming MIDI signals. This bypasses the quantizer.
- **FREQ's** centre-detent will now expand to 10% making it easier to lock onto correct MIDI pitch. Moving **FREQ** will now transpose Plaits.
- Plaits is a **monophonic** voice. It does not support polyphony outside of 3 engines: **Chords**, **String** and **String Machine**, and those have distinct polyphony behaviours that don't constitute traditional polyphony.
- Plaits' **MODEL** input is now exposed via MIDI CC, letting you select engines from the host. Caution - high CPU.

MARBLES - MIDI BEHAVIOURS

- Marbles will sync to external MIDI clocks until MIDI is disconnected. MIDI START is now the same as pressing **PLAY**.
- Marbles' **CLOK** fader now selects a ratio of the incoming MIDI clock which Marbles can play with. The ratios are: 1/4, 1/3, 1/2, 2/3, 1 (at centre), 3/2, 2, 3 or 4.
- Marbles' **DIV+** and **DIV-** rocker switch controls multiplying ranges of that subdivision: $\circ\bullet\bullet\bullet$ -1/4, $\bullet\circ\circ\bullet$ -1, and $\bullet\bullet\bullet\circ$ -4.
- Wakes compares incoming MIDI note timing with Marbles' clock, and runs Marbles that much ahead, so Marbles' steps land with your notes and your DAW needs no clock offset (leave it at 0).

DEFAULT MIDI CC LIST

PLAITS

3-TIMB, 9-MORP, 14-HARM, 15-FREQ, 20-FMOD $\pm$, 21-TIMB $\pm$, 22-MORP $\pm$, 23-HARM $\pm$, 24-LPGC, 25-LPGD, 26-LEVL, 102-OCTV, 105-MODEL, 64 - Sustain pedal.

MARBLES

t
27-CLOK, 16-BIAS, 85-JITR, 17-LNTH, 86-RNDM,
X
28-SPRD, 29-SKEW, 30-STPS,
Y
18-YSPR, 19-YBIS, 87-YSTP, 104-YDIV
DJVU
31-DJVU, 103-DJLN